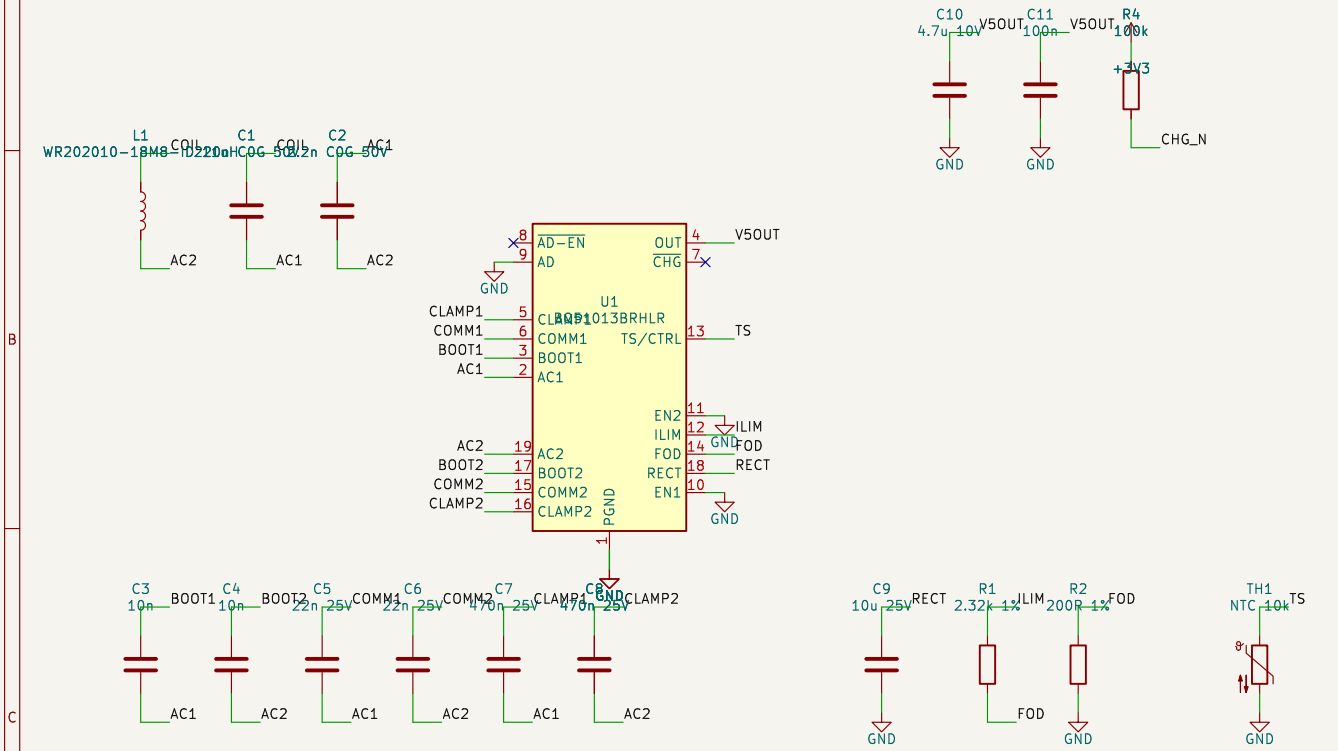
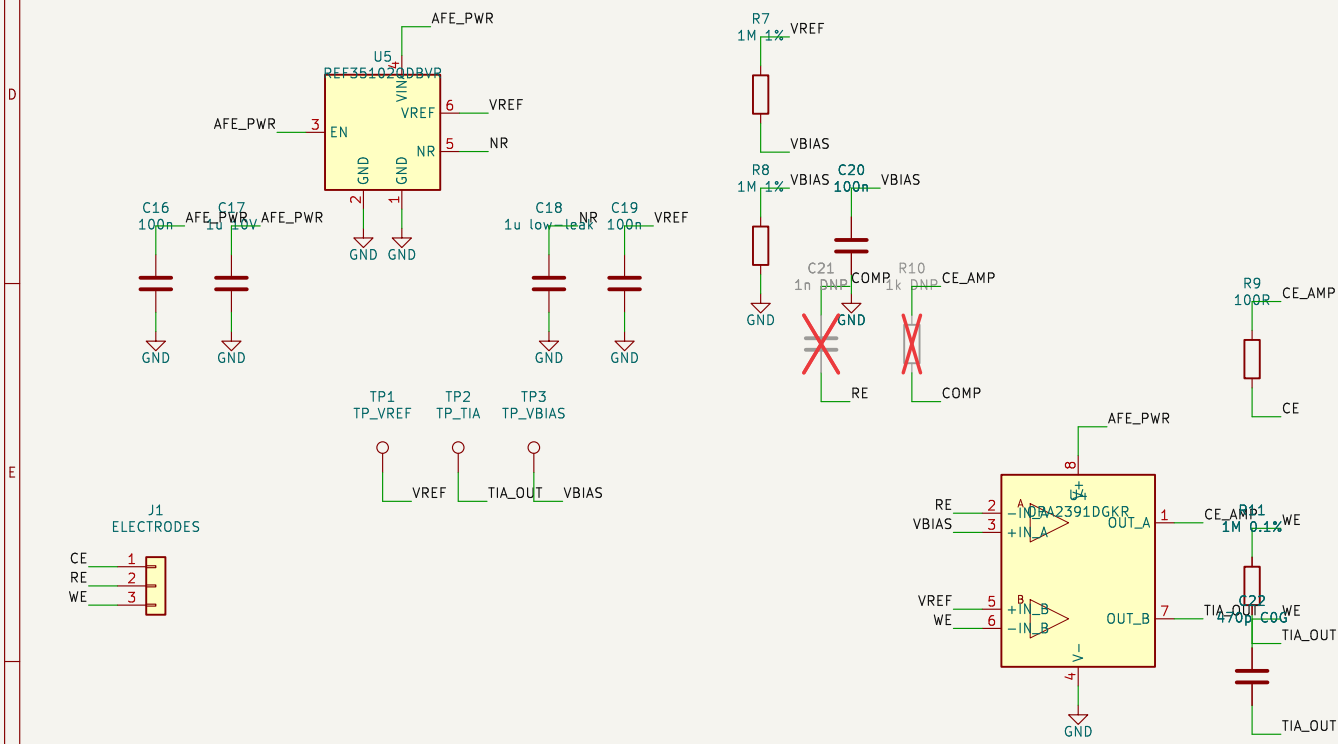


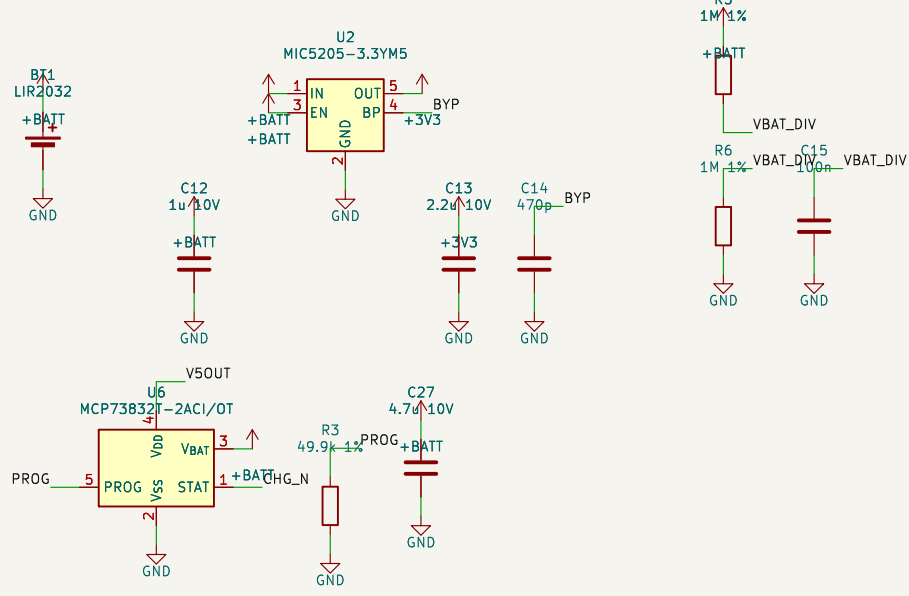
### QI RECEIVER 5V (BQ51013B) + 20mA CHARGER (MCP73832)



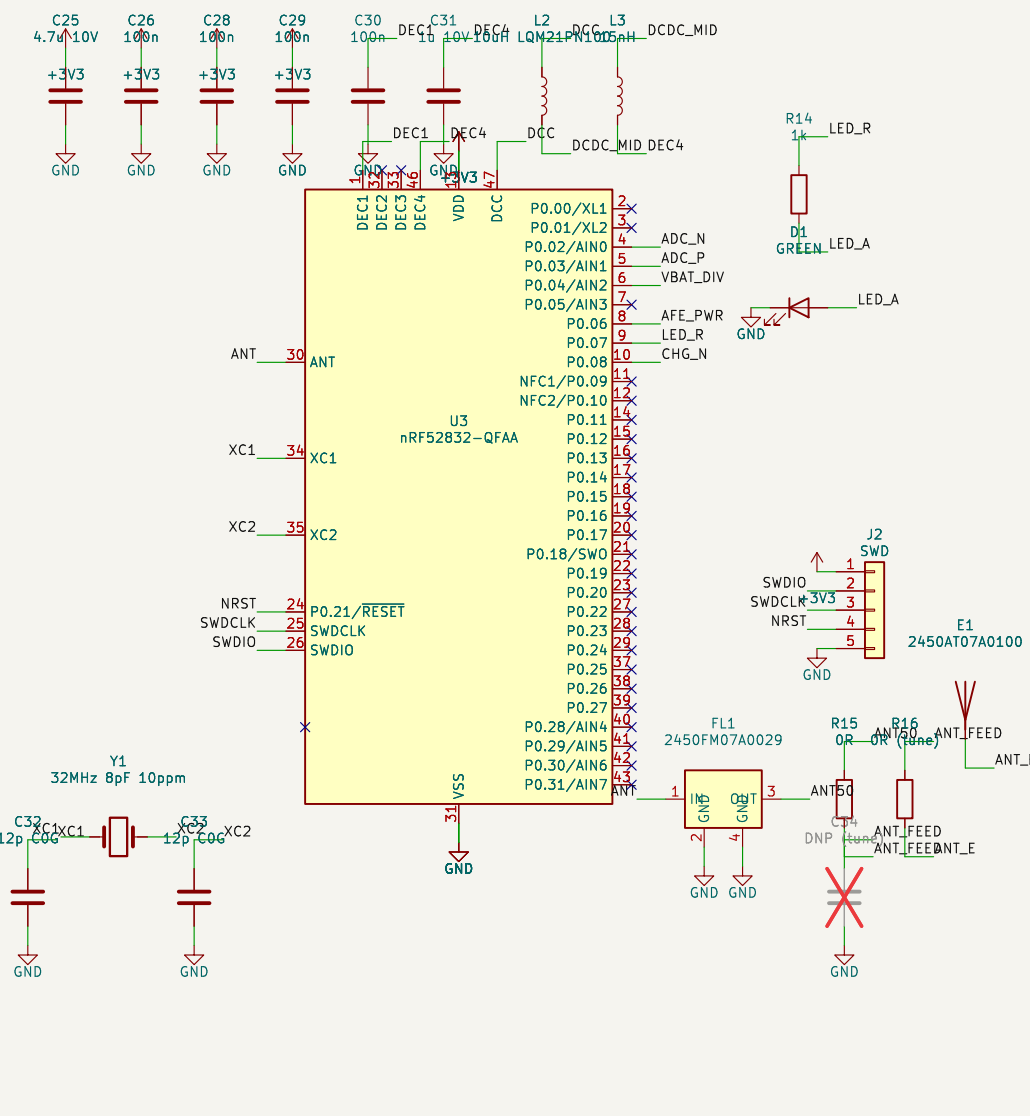
### ANALOG FRONT-END: 2-OP-AMP POTENTIOSTAT (+0.512V)



### BATTERY + 3.3V LDO + VBAT MONITOR



### nRF52832 CHIP-DOWN + 32MHz XTAL + RF FRONT-END + SWD



DISCRETE POTENTIOSTAT – wireless-charged 3-electrode chronoamperometry  
 $E_{cell} = WE - RE = 1.024V - 0.512V = +0.512V$  fixed.  $I_{WE} = (V(ADC\_P) - V(ADC\_N)) / RF$

- DESIGN NOTES / VERIFY BEFORE LAYOUT
- CHARGING: two-stage per TI guidance for small cells. BQ51013B = Qi receiver with regulated 5V (V5OUT). ILIM: RILIM = 250/IMAX = R1+R2 = 2.52k -> IMAX 100mA (HW limit 120mA). MCP73832-2 (4.2V) charges the LIR2032: ICHG = 1000/RPROG = 1000/49.9k = 20mA (0.5C), termination -7.5% = 1.5mA, open-drain STAT -> CHG\_N (100k pullup to 3V3). This replaces the earlier BQ51050B direct-charge (TI: not recommended <200mA).
  - RESONANT TANK: L1 11uH (TDK WR202010-18M8-ID, 20x20x1.0mm, DCR 350mR).  $C1 = 1/(L*(2\pi*100kHz)^2) = 230n$  -> 220n COG (fs 102kHz). C2 -> fd -1MHz -> 2.2n COG. Retune both whenever the coil changes. Small coil = tighter TX alignment; fine at our 0.13W. FL1 (Johanson 2450FM07A0029) replaces the discrete LC match for nRF52 single-ended ANT. R15/C34/R16 = full tee tuning slots per Johanson (series-shunt-series; fit OR/DNP/OR initially).
  - CHIP-DOWN RF: copy Nordic's QFN48 reference layout verbatim (trace geometry included). E1 = 0402 micro antenna; keep the datasheet ground keepout exactly; expect finer tuning. DEC2, DEC3 unconnected per QFN48 reference; no 32k crystal -> firmware uses LFRC synth.
  - TS/CTRL: 10k NTC against the LIR2032 holder cuts wireless power if the cell heats (plain 10k = sense defeated, still a valid safe strap). EN1=EN2=GND; wireless enabled.
  - AFE\_PWR is driven by nRF GPIO P0.06 set to high-drive. Load = REF35+OPA2391 ~ 60uA. Power the AFE off between runs to avoid electrode polarization.
  - WE net: guard ring driven from VREF, no soldermask opening at TIA input, clean flux.
  - RF/CF options: 1M/470p (default, +2.1uA/-0.9uA FS), 5.1M/100p (427/181nA), 10M/47p (217/92nA). rLohm + oversample are runtime settings in firmware; recalibrate after a swap.
  - R10/C21 control-amp compensation are DNP; fit only if the CE-RE loop oscillates.
  - LIR2032 has no protection circuit; firmware must System-Off below 3.0V (VBAT\_DIV, AIN2).
  - SAAD: differential P=AIN1(ADC\_P) N=AIN0(ADC\_N), gain 1/2..4, oversample 64x default.
  - XL1/XL2 (P0.00/P0.01) free: no 32.768k crystal fitted. NFC pins unused -> plain GPIO/NC.